

Computer Science & IT

Data Structure & Programming



Array-02

Lecture No. 02



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Recap of Previous Lecture



Topic

1-D array

Topic

2-D array Row major order

Topic

2-D array Column major order

Topic

Topic

Topics to be Covered



Topic

LTM - Row major ordering

Topic

LTM - column major order

Topic

3-D matrix

Topic

Tridiagonal Matrix (optimal)

Topic

Type of Matrix	Ordering	Address of A[i][j]
2-D Array	Row major Order	$BA + [(i - lb_1) \times (ub_2 - lb_2 + 1) + (j - lb_2)] \times S$
2-D Array	Column Major Order	$BA + [(j - lb_2) \times (ub_1 - lb_1 + 1) + (i - lb_1)] \times S$



Topic: LTM Row Major Order

$A[1][1]$	a_{12} 0	a_{13} 0	a_{14} 0
$A[2][1]$	$A[2][2]$	a_{23} 0	a_{24} 0
$A[3][1]$	$A[3][2]$	$A[3][3]$ ✓	a_{34} 0
$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$

$$A[i][j] = 0 \quad j > i$$

Row will be stored one after another
(Non zero elements only)

Row. 1 Row. 2 Row. 3 Row. 4

$A[1][1]$	$A[2][1]$	$A[2][2]$	$A[3][1]$	$A[3][2]$	$A[3][3]$	$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$
1000	1002	1004	1006	1008	1010	1012	1014	1016	1018

A[1][1]	0	0	0
A[2][1]	A[2][2]	0	0
A[3][1]	A[3][2]	A[3][3]	0
A[4][1]	A[4][2]	A[4][3]	A[4][4]

In 4th Row, we are in 3rd column
 No. of element crossed: $3-1=2$

Address of A[4][3]

No. of Rows crossed: $4-1=\underline{3}$

No. of element in each Row

Row - 1	—	1	} total = 6
		+	
Row - 2	—	2	
		+	
Row - 3	—	3	

total: $6+2=8$

Add. A[4][3] = $1000+8 \times 2$
 $= 1016$



Topic: LTM Row Major Order

A[1][1]	0	0	0
A[2][1]	A[2][2]	0	0
A[3][1]	A[3][2]	A[3][3]	0
A[4][1]	A[4][2]	A[4][3]	A[4][4]

Address of $A[i][j]$

BA is Base Address, S is size

$$BA + [1 + 2 + 3 + \dots + (i-1) + (j-1)] \times S$$

$$BA + \left[\frac{i(i-1)}{2} + (j-1) \right] \times S$$

← Lower Bound
= 1



Topic: LTM Row Major Order



$A[1...100][1...100]$ is lower triangular matrix. Base address is 1000 and each element occupies 2-Bytes of space. What is the address of $A[50][50]$ in row major order ?

$$BA + \left[\frac{i(i-1)}{2} + (j-1) \right] \times \text{Size}$$

$$1000 + \left[\frac{50 \times 49}{2} + 49 \right] \times 2 = 1000 + [25 \times 49 + 49] \times 2$$
$$= 1000 + [1225 + 49] \times 2$$

$$1000 + 1274 \times 2$$
$$1000 + 2548 = 3548$$



Question : Lower Triangular Matrix

#Q $A[-3..10][3..16]$ LTM RMO, $BA = 200$, Size = 2B

Address $A[8][5]$ in RMO

Address of $A[8][5]$
 $+4 \left(\begin{array}{c} 6 \\ A[12][9] \end{array} \right) \left(\begin{array}{c} -2 \\ \end{array} \right)$

Bound Adjustment

$A[-3..10][3..16]$

$+4 \left(\begin{array}{c} -2 \left(\begin{array}{c} \end{array} \right) \end{array} \right) \left(\begin{array}{c} \end{array} \right)$
 $A[1..14][1..14]$

$$200 + \left[\frac{6 \times 11}{2} + 2 \right] \times 2$$

$$200 + 68 \times 2$$

$$200 + 136 = \boxed{336}$$



Topic : Lower Triangular Matrix: Column Major Order

$A[1][1]$	0	0	0
$A[2][1]$	$A[2][2]$	0	0
$A[3][1]$	$A[3][2]$	$A[3][3]$	0
$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$

Column - 1				Column - 2			Column - 3	Column - 4	
A[1][1]	A[2][1]	A[3][1]	A[4][1]	A[2][2]	A[3][2]	A[4][2]	A[3][3]	A[4][3]	A[4][4]
1000	1002	1004	1006	1008	1010	1012	1014	1016	1018



Topic : Lower Triangular Matrix: Column Major Order

A[1][1]	0	0	0
A[2][1]	A[2][2]	0	0
A[3][1]	A[3][2]	A[3][3]	0
A[4][1]	A[4][2]	A[4][3]	A[4][4]

In 3rd column we are in 4th Row
No. of element crossed = 1
(i - j) elements crossed.

Address of A[4][3]

We are in 3rd column

No. of columns completed: $3 - 1 = 2$

No. of element completed:

Column-1 $\rightarrow 4$
Column-2 $\rightarrow 3$

$$\text{total} = 7 + 1 = 8$$

$$\text{Add} : 1000 + 8 \times 2 = 1016$$



Topic : Lower Triangular Matrix: Column Major Order

$A[1][1]$	0	0	0
$A[2][1]$	$A[2][2]$	0	0
$A[3][1]$	$A[3][2]$	$A[3][3]$	0
$A[4][1]$	$A[4][2]$	$A[4][3]$	$A[4][4]$

No. of element crossed
jth Row ith column
(i-j)

Address of $A[i][j]$

(j-1) columns completed

Column 1 $\longrightarrow n - 0$
+

Column 2 $\longrightarrow n - 1$
+

Column 3 $\longrightarrow n - 2$
+

⋮

Column j-1 $\longrightarrow n - (j-2)$

$$= n(j-1) - \frac{(j-1)(j-2)}{2}$$

$$n + n - 1 + n - 2 + n - 3 + \dots + n - (j - 2)$$

AP series

$$\frac{n}{2}(a + l)$$

No. of terms: $(j - 1)$

first term: n

Last term: $n - (j - 2)$

$$\frac{(j - 1)}{2} [n + n - (j - 2)]$$

$$\frac{(j - 1)}{2} [2n - (j - 2)]$$

$$n(j - 1) - \frac{(j - 1)(j - 2)}{2}$$

LTM CMO

$$BA + \left[n(j-1) - \frac{(j-1)(j-2)}{2} + (i-j) \right] \times S$$

LTM RMO

$$BA + \left[\frac{i(i-1)}{2} + (j-1) \right] \times S$$



Topic : Upper Triangular Matrix

a_{11}	a_{12}	a_{13}	a_{14}
0 a_{21}	a_{22}	a_{23}	a_{24}
0 a_{31}	0 a_{32}	a_{33}	a_{34}
0 a_{41}	0 a_{42}	0 a_{43}	a_{44}

$$A[i][j] = 0 \text{ if } i > j$$

UTM :- Column major order, Address of $A[i][j]$

$$BA + [1 + 2 + 3 + \dots + (j-1) + i-1] \times \text{Size}$$

$$BA + \left[\frac{j(j-1)}{2} + (i-1) \right] \times \text{Size}$$



Topic : Upper Triangular Matrix

a_{11}	a_{12}	a_{13}	a_{14}
0	a_{22}	a_{23}	a_{24}
0	0	a_{33}	a_{34}
0	0	0	a_{44}

Address of $A[i][j]$

UTM Row major order, i^{th} Row, j^{th} column

$$BA + [n + (n-1) + (n-2) + \dots + n - (i-2)]$$

$$+ (j-i) \times \text{Size}$$

$$BA + \left[n(i-1) - \frac{(i-1)(i-2)}{2} + (j-i) \right] \times \text{Size}$$

LTM/UTM

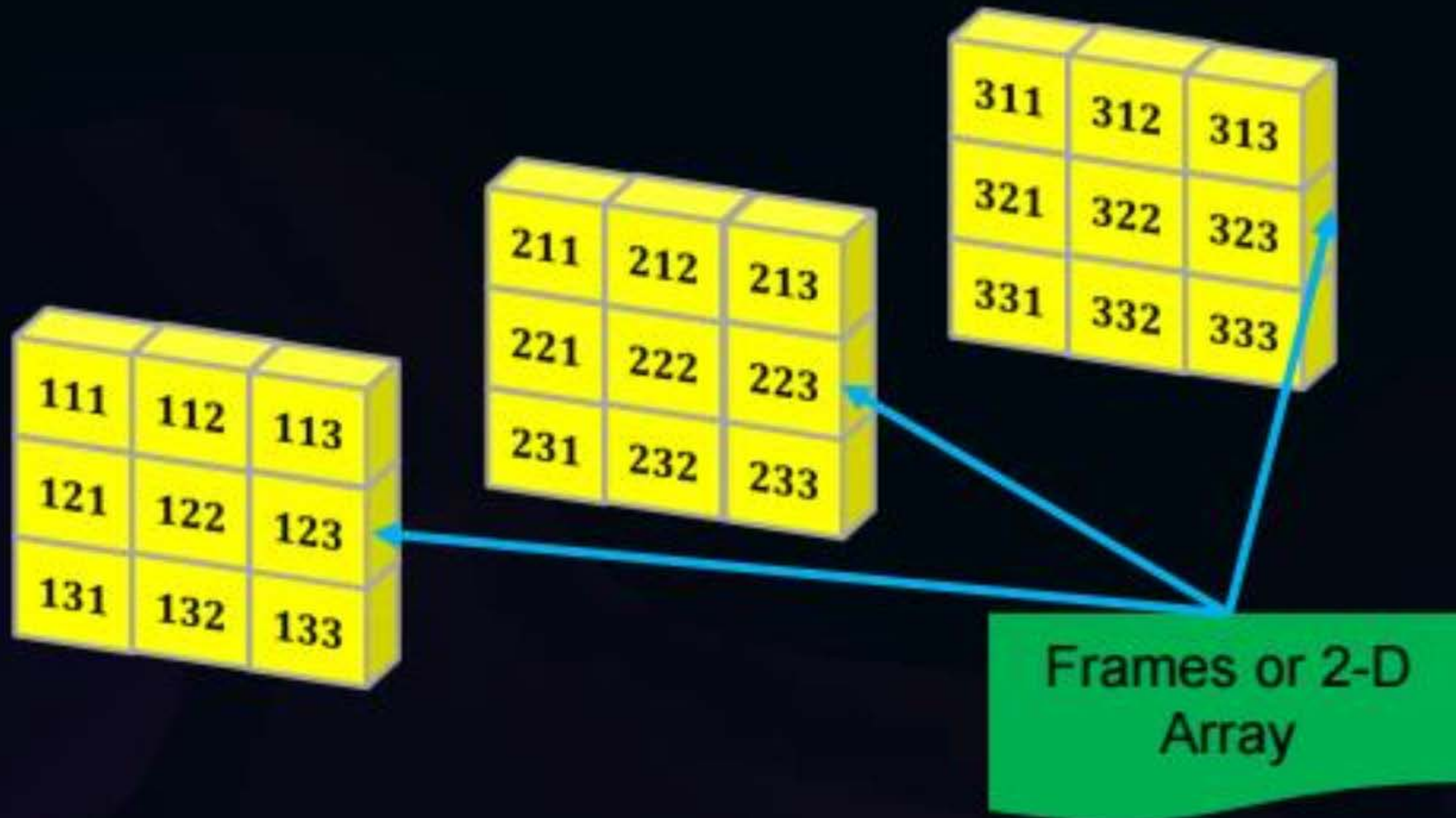
Type of Matrix	Ordering	Address of A[i][j]
Lower Triangular Matrix	Row major Order	$BA + \left[\frac{l \times (l-1)}{2} + (j-1) \right] \times S$
Lower Triangular Matrix	Column Major Order	$BA + \left[n(j-1) - \frac{(j-1) \times (j-2)}{2} + (i-j) \right] \times S$
Upper Triangular Matrix	Row major Order	$BA + \left[n(i-1) - \frac{(i-1) \times (i-2)}{2} + (j-i) \right] \times S$
Upper Triangular Matrix	Column Major Order	$BA + \left[\frac{j \times (j-1)}{2} + (i-1) \right] \times S$



3-D Array



$A[1..3][1..3][1..3]$



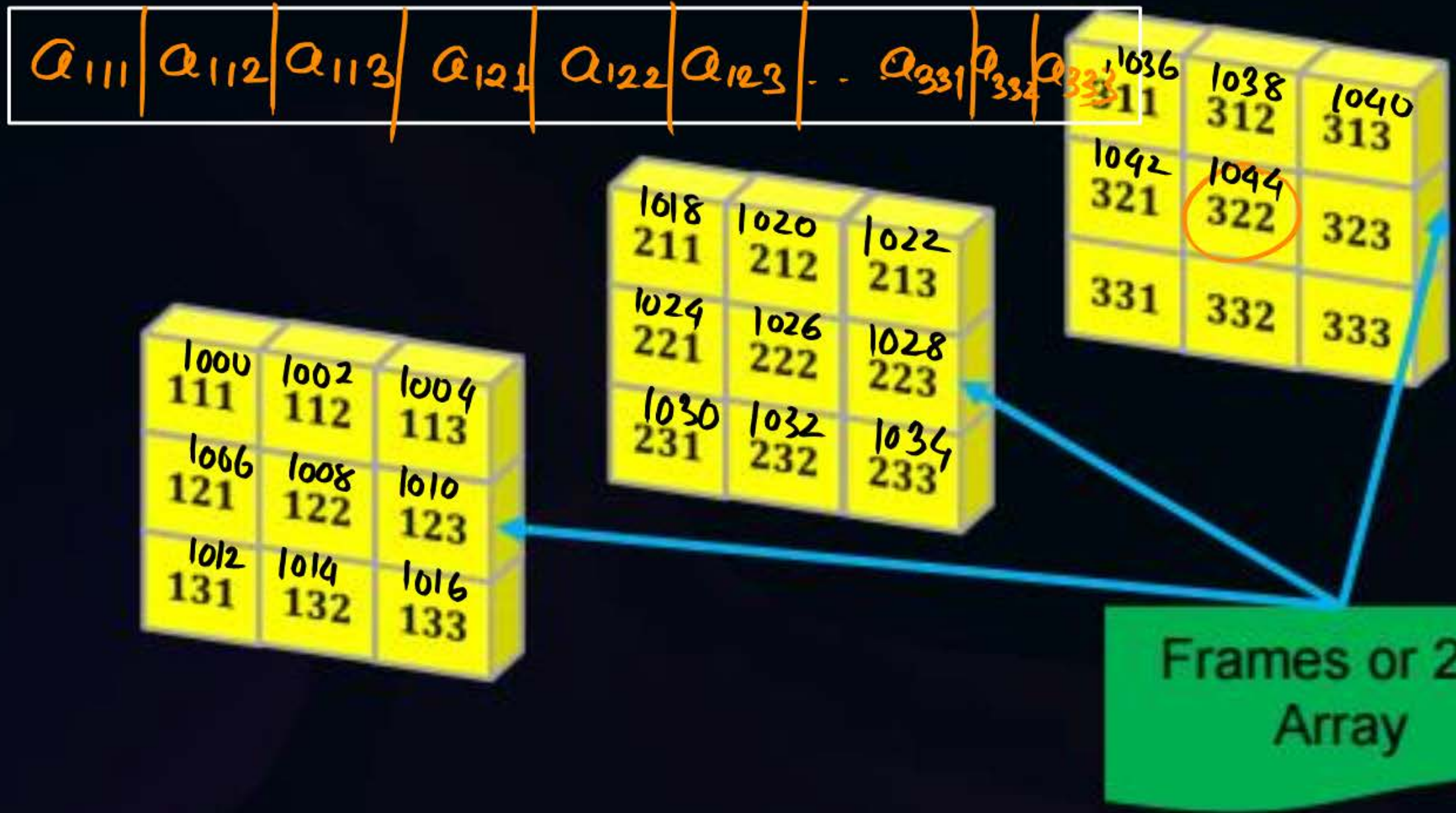


3-D Array

C-Language

$A[1..3][1..3][1..3]$

$A[3][2][2]$
↑





3-D Array



BA - 1000 Size = 2B

$$\begin{array}{c} A[3][2][2] \\ \uparrow \quad \uparrow \quad \uparrow \\ \text{frame} \quad \text{Row} \quad \text{column} \end{array}$$
$$1000 + \left[(3-1) \times 3 \times 3 + (2-1) \times 3 + (2-1) \right] \times \text{Size}$$

$$1000 + [18 + 3 + 1] \times 2$$

$$1000 + 44 = 1044$$



3-D Array



$A[LB_1..UB_1][LB_2..UB_2][LB_3..UB_3]$

Address of $A[i][j][k]$

$$BA + \left[\underbrace{(i - LB_1)}_{\substack{\text{\# of frames} \\ \text{arranged}}} \times \underbrace{(UB_2 - LB_2 + 1)}_{\substack{\text{\# of elements} \\ \text{in Row}}} \times \underbrace{(UB_3 - LB_3 + 1)}_{\substack{\text{\# of elements in} \\ \text{column}}} + \right.$$

$$\left. \underbrace{(j - LB_2)}_{\substack{\text{\# of Rows} \\ \text{arranged}}} \times \underbrace{(UB_3 - LB_3 + 1)}_{\substack{\text{\# of element in} \\ \text{Column}}} + (k - LB_3) \right] \times \text{Size}$$



Question : 2-D Array

#Q

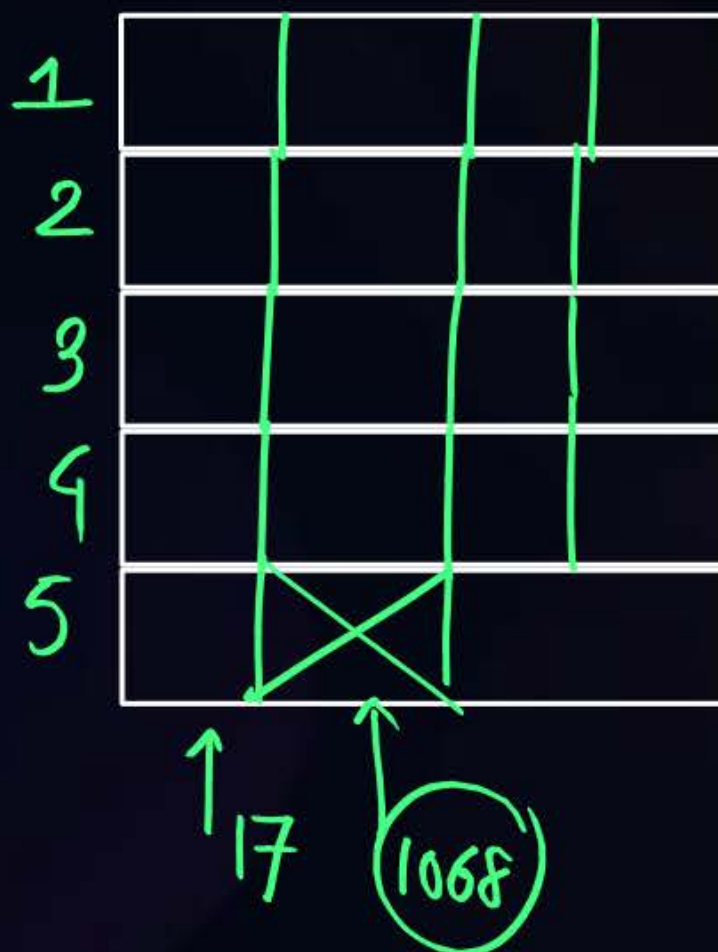
Suppose you are given with 2-D Array A, where array is arranged in row major order and each row consists of 4 elements. Let the base address is 1000 size of each element is 4B. An element $A[i][j]$ is stored in 1068 then what is value of $i+j$ [7]. (Lower bound is 1)

element 1068

element address: 1000 to 1067 = 68B

No. of element arranged $\frac{68}{4} = 17$

$A[5][2]$





Question : 2-D Array

#Q

Suppose you are given with 2-D Array A, where array is arranged in row major order and each row consists of 7 elements. Let the base address is 1000 size of each element is 2B. An element $A[i][j]$ is stored in 1068 then what is value of $i+j$ [12]. (Lower bound is 1)

1000 to 1067 - 68 Bytes

$$\frac{34}{7} \quad \underline{Q: 4} \quad \underline{R: 6}$$

$$\text{No. of elements} = \frac{68}{2} = \underline{34}$$

$$\text{Row} = 4 + 1 = 5$$

$$\text{column} = 6 + 1 = 7$$

$$A[5][7]$$



Topic : Question



#Q A[1...100]1...100] is lower triangular matrix. Base address is 1000 and each element occupies 2-Bytes of space if the address of A[40][j] in row major order is 2578 then value of j is ____?

$$2578 = 1000 + \left[\frac{40 \times 39}{2} + (j-1) \right] \times 2$$

$$\frac{1578}{2} = \left[20 \times 39 + (j-1) \right]$$

$$789 = 780 + j - 1$$

$$j - 1 = 9 \Rightarrow j = 10$$



Topic : Question



#Q A lower-triangular matrix $B[1...120][1...120]$ is stored in column-major order. The base address = 5000, and each element occupies 4 bytes. If the address of $B[x][15]$ is 11376, find the value of x .

$$\underset{5000}{11376} = BA + \left[n(j-1) - \frac{(j-1)(j-2)}{2} + (i-j) \right] \times 2$$

$$\frac{6376}{4} = \left[120 \times 14 - \frac{14 \times 13}{2} + (x - 15) \right]$$

$$1594 = 1680 - 91 + x - 15$$

$$\boxed{x = 20}$$



Topic : Question

#Q A[1...100][1...100][1...100]. Base address is 1000 and each element occupies 2 Bytes of space. What is the address of A[50][49][50] in row major order?

$$1000 + [(50-1) \times 100 \times 100$$

$$+ (49-1) \times 100$$

$$+ (50-1)] \times 2$$

$$1000 + [490000 + 4800 + 49] \times 2$$

$$1000 + 494849 \times 2 = \boxed{990698}$$
$$1000 + 989698 = \boxed{990698}$$

(A) 900896

(B) 990896

(C) 890896

(D) None of the above ✓



Topic: Practice Problem

Q Consider the natural numbers from are stored in a 2D array $\text{arr}[1 \text{ to } 40][1 \text{ to } 30]$.

Suppose, the elements are stored in row-major orde. If $A[i][j]$ consists of 1030 then value of

$i+j$ is 45

1	1	2	3	4	5	...	30
2	31						60
...							
34							
35	1021	1022	1023	.	...	1030	

$$\frac{1030}{30} \quad Q \ 34 \ R: 10$$

Stack~

THANK - YOU